UU-P-233b
October 15, 1962

SUPERSEDING
Int. Fed. Spec. UU-P-00233a(GSA-FSS)
March 23, 1962 and
Fed. Spec. UU-P-231a (IN PART)
May 24, 1954 (See 6.3.)

FEDERAL SPECIFICATION

PAPER, DUPLICATING, MASTER, SPIRIT-PROCESS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers white master paper in sheet form for use in spirit-process duplicating operations (see 6.1).

1.2.1 *Type and grade.* The master duplicating paper shall be of but one type and grade.

1.2.2 *Size and trim — sheets.* The master duplicating paper shall be furnished in sheets of the size(s) ordered and shall be flat, guillotine-trimmed square on four sides with smooth edges. The sheets shall not be undersize nor more than 1/16 inch oversize in either direction. (See 6.2.)

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Specifications:

O-D-825—Duplicating Fluid, Direct Process, Spirit-Type.

UU-P-150—Paper, Carbon, Duplicating Machine, Spirit-Process.

UU-P-232—Paper, Duplicating, Copy, Liquid Process.

PPP-B-601—Boxes, Wood, Cleated-Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636—Box, Fiberboard.

PPP-T-45—Tape; Gummed, Paper, Reinforced.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-P-116—Preservation, Methods of.

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Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications, of the issues in effect on date of invitation for bids, form a part of this specification:

Technical Association of the Pulp and Paper Industry Publications:

T 401 m—Fiber Analysis of Paper and Paperboard

T 410 m—Basis Weight of Paper and Paperboard.

T 411 m—Thickness and Density of Paper.

T 459 m—Wax Test for Surface Strength of Paper.

T 479 sm—Smoothness of Printing Paper.

(Copies of T.A.P.P.I. testing methods may be obtained from the Technical Association of the Pulp and Paper Industry, 360 Lexington Avenue, New York 17, N. Y.)

3. REQUIREMENTS

3.1 Material. The master duplicating paper shall be made from such material as to insure compliance with the requirements of this specification.

3.2 Color, formation, and cleanliness. The color of the master duplicating paper shall be white. The master duplicating paper shall be at least equal to the Government standard comparison sample with respect to color, formation, and cleanliness (see 6.4).

3.3 Physical characteristics. The master duplicating paper shall have the physical characteristics specified below (see 4.4.1).

Paper stock—Free from groundwood pulp.

Weight—25 x 38, 1,000 sheets, 120 pounds.

Thickness—Average, 0.0030 inch.

3.3.1 *Weight tolerance.* A plus or minus tolerance of 5 percent will be allowed for weight (1,000-sheet) of the paper.

3.3.2 *Thickness tolerance.* A plus or minus tolerance of 0.0005 inch will be allowed for thickness of the master duplicating paper. Individual thickness of sheets shall be uniform and shall not vary more than 0.0004 inch from one edge to the other.

3.4 Coating. The master duplicating paper shall be coated two sides, the coating shall be uniform (see 3.3.2), and shall be free from mottle and any particles which will pick or lift under normal use conditions. (See table I.)

3.5 Finish. The master duplicating paper shall have a finish (smoothness), average (each side), of not less than 500 seconds when tested as specified in 4.4.1.

3.6 Surface strength. The master duplicating paper shall withstand a No. 4A Dennison wax test on both sides of the sheet without disturbance of the coating when tested as specified in 4.4.1.

3.7 Curl. The master duplicating paper shall lie flat with no tendency to curl (see 4.4.2).

3.8 Resistance to duplicating fluid. The master duplicating paper shall not be affected by the duplicating fluid specified in 4.4.3.3 to such an extent that it interferes with satisfactory performance when tested as specified in 4.4.3.

3.9 Grain direction. The grain direction of the master duplicating paper shall parallel the long direction of the sheet.

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3.10 Duplicating quality (performance). Each side of the master duplicating paper when used with carbon paper conforming to UU-P-150, type II duplicating fluid conforming to O-D-825, and copy paper conforming to UU-P-232, shall be capable of reproducing legible copies, with sharp character impressions from the same master when tested as specified in 4.4.3, as indicated below:

Carbon paper (color)	*Number of copies (minimum)*
Purple	300
Blue	300
Black	100
Other colors	250

Faithfulness of reproduction of character shall remain unimpaired throughout the number of copy reproductions indicated above for the respective colors in respect to continuity and shape, using either of the typewriter pressure settings specified in 4.4.3.2.

3.11 Preprinted masters. Preprinting on master duplicating paper sheets, when required, shall be as specified in the invitation for bids, contract, or order (see 6.2).

3.12 Workmanship. The master duplicating sheet shall be clean and free from pinholes and any other defects which affect the appearance or which might impair the satisfactory performance of the master paper (see table I).

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier's responsibility. Unless otherwise specified (see 6.2), the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations, and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot of master duplicating paper shall consist of all paper of one size offered for delivery at one time.

4.3 Sampling and inspection.

4.3.1 *Sampling for visual examination* shall be performed using the procedures prescribed by MIL-STD-105. The normal inspection levels (sample sizes) and acceptable quality levels (AQL's) expressed in defects per hundred units, shall be as indicated below:

For examination	*Inspection level*	*Acceptable quality levels*
4.3.1.1	L-6	4.0
4.3.1.2	L-4	2.5
4.3.1.3	L-2	Average
4.3.1.4	L-4	40

4.3.1.1 *Shipping containers.* Packed, sealed, and marked shipping containers shall be examined for compliance with section 5 of this specification. The sample size shall be examined for freedom from the defects listed in table I. The sample unit shall be one shipping container.

4.3.1.2 *Reams.* The sample reams shall be examined for freedom from the defects listed under "Reams" in table I. Approximately the same number of reams shall be taken from each shipping container sampled in accordance with 4.3.1.1. The sample unit shall be one wrapped and labelled ream.

4.3.1.3 *Ream count.* Sample reams shall each be counted to determine compliance with ream count requirements. The sample unit shall be one ream.

4.3.1.4 *Sheets.* Ten consecutive sheets of master duplicating paper shall be taken

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from each ream examined in accordance with 4.3.1.2. Not fewer than 75 sheets or equivalent area, shall be examined for minor defects. The sample unit shall be one cut size sheet or equivalent area.

4.3.2 *Sampling for tests.* Sample sheets of master duplicating paper shall be taken from the sample reams selected in accordance with 4.3.1.2. Not less than 10 consecutive sheets shall be selected from each ream.

4.3.3 *Examination.* Each sample selected for examination in accordance with 4.3 shall be examined for defects as listed in table I.

TABLE I.—*Classification of defects*

Examine	Defects
Major:	
Preparation for delivery:	
Shipping containers.	Wrong box style, type, and class, and otherwise do not conform to applicable requirements of PPP-B-636.
	Container not closed, flaps do not meet within 1/4 inch.
	Glue does not cover at least 50 percent of area of contact of inner and outer flaps.
	Box wall punctured, torn.
	Score lines torn or ripped.
	Wrong box weight or construction.
	Top and bottom seams not closed with 3-inch reinforced tape (if applicable).
Marking (containers).	Marking missing, illegible, incomplete, or erroneous.
Unit packages (reams):	
Marking (or labeling).	Marking (or labeling) missing, illegible, incomplete, or erroneous.
Wrapping.	Not wrapped as specified.
Count (ream average shall not be less than that specified).	Ream contains fewer sheets than specified.
Paper, master:	
Color, formation, and cleanliness.	Do not compare favorably with the Government standard comparison sample. Variation within ream. Variation within order.
Dimensions and trim.	Not size specified within (oversize) tolerance. Undersize. Not square. Not grain direction specified in order. Not guillotine trimmed.
Repetitive sheet defects within reams (more than 2½ percent of sheets affected).	Welts; wrinkles; creases; tear; curl; grainy sheets; uneven, wavy, or ragged edges; crushed corners.
Coating (both sides).	Coating mottled; not uniform—gritty, lumpy, stippled, ribbed, stripped. Loose particles—lint, dust, fuzz, flakes of coating.
Workmanship.	Not as specified herein.
Minor:	
Paper, master:	
Cleanliness and paper defects.	Sheets do not compare favorably with the Government standard comparison sample with respect to: Shives; dirt specks; knots; oil, grease, size, dye, or slime spots; slime holes, pin holes, or craters; fish eyes; shiners; lumps; calendar blackening, or cuts, scale, or scabs.

4.4 Tests.

4.4.1 *Physical characteristics and surface strength of paper.* Sample sheets of master duplicating paper in accordance with 4.3.2 shall be tested as indicated in table II.

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TABLE II

Tests	T.A.P.P.I. test method number
Fiber identification and quantitative determination	T 401 m
Weight	T 410 m
Thickness	T 411 m
Finish (smoothness) (Bekk tester)	T 479 sm
Surface strength of paper, wax test for	T 459 m[1]

[1] For additional information on the wax test method used, consult the methods of the Technical Association of the Pulp and Paper Industry, 360 Lexington Avenue, New York 17, New York.

4.4.2 *Curl.* During the performance test (4.4.3), any curl of the master duplicating paper which interferes with satisfactory copy reproduction shall be considered as failing to meet the requirements of this specification.

4.4.3 *Performance.*

4.4.3.1 *Apparatus.* The preparation of the duplicating master shall be made on an electric typewriter with pica type and in good operating condition. The typewriter shall be equipped with a typewriter ribbon not more than 0.0050 inch thick and a platen having a Shore durometer reading of between 89 to 93 or an equivalent platen designed for the preparation of duplicating masters on the machine being used. The type faces shall be thoroughly cleaned before each test.

4.4.3.2 *Preparation of master.* The master paper shall be placed in the typewriter with a sheet of carbon paper conforming to UU-P-150 with the carbon paper placed carbon-up, under and against the master sheet (see 3.10). The master shall be typed with the entire keyboard, small letters, capitals, numerals, and other characters twice in unrelated order, with the stroke of the typewriter set at one-fourth of its full pressure. The same typing shall be repeated with the stroke of the typewriter set at one-half of its full pressure.

4.4.3.3 The typed master prepared as specified in 4.4.3.2 shall be placed on a spirit-type duplicating machine, and using type II duplicating fluid conforming to O-D-825, together with copy paper conforming to UU-P-232, the specified number of copies (see 3.10) shall be reproduced. The reproduction shall then be examined for compliance with 3.4 and 3.10.

5. PREPARATION FOR DELIVERY

Levels of packaging and packing protection specified hereinafter shall be as defined in Fed. Std. No. 102 (for civil agencies only).

5.1 **Packaging.** Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 *Unit (1-ream) packages.*

5.1.1.1 *Level A.* Unless otherwise specified, level A unit packaging shall be as follows: Unit packages of master duplicating paper shall each consist of one ream (500 sheets). Each ream shall be compactly wrapped in good quality kraft paper, having a minimum basis weight of 40 pounds, and packaged in accordance with MIL-P-116, method IC-2.

5.1.1.2 *Level B.* Unit packages of master duplicating paper shall each consist of one ream (500 sheets). Each ream shall be compactly wrapped in good quality kraft paper having a minimum basis weight of 40 pounds. Wrapping shall be sealed with an adhesive or gummed paper tape. Adhesive, if used, shall be applied in a manner so that no contact of adhesive is made with the master duplicating paper being wrapped.

5.1.1.3 *Level C.* Unless otherwise specified, master duplicating paper shall be packaged in accordance with standard commercial practice.

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5.2 **Packing.** Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Level A.* Unless otherwise specified, level A packing shall be as follows: Master duplicating paper packaged as specified in 5.1.1.1, shall be packed in boxes conforming to PPP-B-621, class 2, style optional; PPP-B-601, overseas type, style optional; or PPP-B-636, class 2, V3c, style optional. The gross weight of wood boxes shall not exceed 200 pounds; of fiberboard boxes, 70 pounds. Wood boxes shall be strapped in accordance with the appendix to the applicable box specification. Fiberboard boxes shall be reinforced with tape in accordance with the appendix to the box specification.

5.2.2 *Level B.* Ten reams of master duplicating paper, packaged as specified in 5.1.1.2, shall be packed in new fiberboard boxes conforming to style RSC, type I or II, class 1 of PPP-B-636, except that bursting strength shall be a minimum of 350 p.s.i. for weights in excess of 65 pounds and 275 p.s.i. for weights up to and including 65 pounds. Sealing of the boxes shall be accomplished by applying adhesive to the inner and outer flaps over an area not less than 50 percent of area of contact. The seams, top and bottom, shall be closed in accordance with the appendix to PPP-B-636, or, alternatively, by means of 3-inch reinforced gummed tape conforming to PPP-T-45.

5.2.3 *Level C.* Unless otherwise specified, unit packages shall be packed in accordance with standard commercial practice. Shipping containers shall comply with the carrier rules and regulations applicable to the mode of transportation.

5.3 **Marking or labeling.**

5.3.1 *Civil agencies.* In addition to any special marking specified in the invitation for bids, contract, or order (see 6.2), marking shall be as follows:

5.3.1.1 *Unit (1-ream) packages.* Unless otherwise specified, marking of unit (1-ream) packages shall be in accordance with Fed. Std. No. 123. Each wrapped unit package shall be marked or labeled on one end with the following information:

(a) Applicable Federal Stock Number.

(b) Federal Item Identification, including specification classification of the commodity (size, etc.).

(c) Quantity of units contained therein (e.g.: UNIT—REAM/QUANTITY-1).

(d) Name of contractor.

(e) Contract and/or purchase order number.

5.3.1.2 *Shipping containers.* Unless otherwise specified, each shipping container shall be marked in accordance with Fed. Std. No. 123, including specification classification of the commodity (size, etc.).

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, marking of interior packages and shipping containers shall be in accordance with MIL-STD-129.

5.4 **Standard pack.** The standard pack requirements for the commodity covered by this specification, including marking for shipment, shall be as specified in 5.1.1.2, 5.2.2, and 5.3.1 through 5.3.1.2. (see 6.2.1.)

6. NOTES

6.1 **Intended use.** The master duplicating paper covered herein is intended for use on spirit-process duplicating machines. The paper sheets are intended for use as masters to be reproduced in spirit-process duplicating operations. (See 3.10.) Reproduction by spirit-process duplicators can be made from a suitable, typed, written, or drawn original (carbon-impression backed) master. Pencils (preferably hard lead), ball-point pens, and standard commercial typewriters of the manual and electrical types are commonly used in preparing the masters.

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6.2 **Ordering data.** Purchasers should exercise any desired options offered herein (see 5.1.1.1, 5.1.1.3, 5.2.1, 5.2.3, 5.3.1.1, and 5.3.1.2) and procurement documents should specify the following (see 3.2):

(a) Title, number, and date of this specification;

(b) Size of sheets required (see 1.2.2).

(c) Whether preprinted master sheets are desired, and if so, the printing required thereon (see 3.11).

(d) Inspection responsibility (*for civil agencies only*) if other than as specified herein; and administrative provisions for inspection records, when applicable (see 4.1).

(e) Level of packaging and packing and any special marking for shipment required (see 5.1, 5.2, and 5.3).

6.2.1 *Standard pack (civil agencies).* The standard pack requirements including marking for shipment, specified in 5.1.1.2, 5.2.2, and 5.3.1 through 5.3.1.2, are intended for use in procurement of Stores Stock replenishments. Procuring officers should use the standard pack provisions when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot, or intermediate storage point for temporary storage, subsequent issue, and/or shipment to eventual user. (See 5.3.2.)

6.3 This specification provides requirements for *master* duplicating paper, superseding those covered in Federal Specification UU-P-231a, Paper, Duplicating, *Copy and Master*, Liquid Process, dated May 24, 1954, and Interim Federal Specification UU-P-00233a (GSA-FSS), Paper, Duplicating, Master, Spirit-Process. (The requirements for *copy* paper included in Federal Specification UU-P-231a have been superseded in Federal Specification UU-P-232, Paper, Duplicating, Copy, Liquid-Process.)

6.4 **Standard comparison sample.** Standard comparison samples may be obtained upon application to the procuring agency issuing the invitation for bids.

6.5 **Transportation data.** The transportation descriptions and minimum weights applicable to the commodity covered herein are:

Rail:

Paper, not otherwise indexed by name, not printed.

Carload minimum weight 36,000 pounds.

Motor:

Paper, not otherwise indexed, not printed.

Truckload minimum weight 36,000 pounds, subject to Rule 34, National Motor Freight Classification.

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